

**Software Installation Guide for
HealthCare Plus Hospital
Online Hospital Management System.**

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1. Introduction.

About this guide.

- This guide provides step-by-step instructions for installing and running the Online Hospital Management System developed for HealthCare Plus Hospital.
- The procedure is intended for a local development or demonstration environment using XAMPP, Apache, PHP and MySQL.

Components Installed.

- XAMPP web-server environment containing Apache, PHP and database services.
- HealthCare Plus application files placed in the XAMPP web directory.
- MySQL relational database containing the hospital system data.
- Required PHP application configuration and third-party API settings, where applicable.
- A supported web browser such as Google Chrome or Microsoft Edge.

Installation Overview.

- The project is designed to run on localhost during development and on a local server or intranet for initial deployment.
- The documented localhost address is 127.0.0.1.
- The system includes patient management, appointment scheduling, billing and payment, medical records/report management, staff/resource management, feedback/support and an AI-powered symptom checker.
- An internet connection may be required for external services such as email or PayPal-related integration.

2. System Requirements.

Component	Minimum/Project Requirement	Recommended
Device	Windows 10	Laptop/Desktop
Processor	Minimum Intel Core i3 class processor	11 th Gen Intel Core i7-1165G7 or equivalent.
RAM	4GB	8GB
Storage	At least 1GB available for the application.	512GB system storage
Display	At least 1024 × 768	1920 × 1080
System	64-bit operating system recommended.	64-bit × 64-based processor

Table 1: System Requirements.

Supporting Operating Systems and Environment.

- Windows 10 or above is the primary environment.
- Modern Linux distributions and macOS are identified as supported client environments in the project documentation; the application is designed for deployment on Windows and Linux servers with minimal reconfiguration.
- Web server: XAMPP with Apache, PHP and MySQL.
- Web browsers: Google Chrome and Microsoft Edge.
- Network: localhost address 127.0.0.1; the report assumes a minimum 5 Mbps connection where internet-dependent features are used.

Software Requirements.

- XAMPP: The report specifies XAMPP SQL Server 2021 – v3.3.0 as the minimum requirement.
- PHP: Server-side language used by the application.
- MySQL: Relational database management system.
- Visual Studio Code or Notepad++: Optional editors used by the project.
- Google Chrome or Microsoft Edge: Browser for accessing the application.

3. Pre-Installation Checklist.

- Complete the following checks before installing the application.
- Administrator access is available on the installation computer.
- The XAMPP installer is available and compatible with the operating system. l The HealthCare Plus project source files are available locally.
- The project database SQL/schema file is available if supplied separately with the source package.
- At least 1 GB of free disk space is available for the basic application; additional space should be reserved for database records and backups.
- Ports required by Apache and MySQL are not occupied by conflicting services.
- A stable network/internet connection is available when external API, email, SMS or payment functionality is being configured.
- A modern browser such as Chrome or Edge is installed.
- The installer has access to the PHP application configuration/connection file so that database credentials can be entered.
- Back up any existing XAMPP project or database before replacing files.

Project File Preparation.

1. Obtain the complete HealthCare Plus project folder from the developer/project package.
2. Keep the project folder unchanged initially so that all PHP, HTML, CSS, JavaScript and supporting files remain together.
3. Locate the database SQL file, if it is included with the project package.
4. Locate the PHP database connection/configuration file and any example configuration files.
5. Keep API credentials separate and do not publish passwords or secret keys in public repositories.

4. Installing Prerequisites.

Installing XAMPP.

1. Download and install the XAMPP version required by the project environment.
2. Run the installer with administrator privileges.
3. Keep Apache and MySQL selected as required components. PHP is included within the XAMPP environment.
4. Complete the installation using the default installation directory unless the project environment requires another location.
5. Open the XAMPP Control Panel after installation.

Starting Required Services.

- Start Apache from the XAMPP Control Panel.
- Start MySQL from the XAMPP Control Panel.
- Confirm that both services report a running state before continuing.

Verification Steps.

- Open the browser and visit: <http://127.0.0.1/>
- If the XAMPP start page appears, Apache is running correctly.
- **Optional Development Tools.**
- Install Visual Studio Code or Notepad++ if source-code editing is required.
- Use Google Chrome or Microsoft Edge for application testing.

5. Database Installation.

This project used MySQL as the relational database to store patient, doctor, appointment, billing and other system data. There are eighteen system tables, including admin, appointment, chatbot_conversations, doctor, doctor_privileges, family_history, feedback, medical_records, medications, patient, reports, schedule, specialities, symptoms_medicines, vital_signs and webuser.

Obtaining the Database Schema.

- Locate the SQL/database file supplied with the HealthCare Plus project.
- If the database file is stored in a database or SQL folder, keep its original filename.
- If no SQL file is supplied, use the database backup/schema delivered with the project rather than creating a new schema manually.

Creating/Importing the Database with phpMyAdmin.

1. Confirm that Apache and MySQL are running in XAMPP.
2. Open a browser and navigate to <http://127.0.0.1/phpmyadmin/>.
3. Create a new database using the database name expected by the project's PHP connection/configuration file. The Final Project Report does not specify a single database name.

4. Select the newly created database.
5. Open the Import tab.
6. Choose the project SQL file.
7. Start the import and wait for phpMyAdmin to report completion.
8. Open the database structure view and verify that the expected tables have been created.

Database Verification.

- Confirm that the core tables are present.
- Confirm that primary and foreign-key relationships supplied by the project schema are intact.
- Verify that test/admin/user records required for demonstration are available, if supplied by the project package.
- Do not modify production or assessment data unless a backup has been created.

6. Application Installation.

1. Locate the XAMPP installation directory.
2. Open the htdocs folder.
3. Copy the complete HealthCare Plus project folder into htdocs.
4. Preserve the project folder structure and filenames.
5. Do not place individual PHP files in unrelated XAMPP folders.

Typical Windows XAMPP web directory: C:\xampp\htdocs\<HealthCare-Plus-project-folder>

Application Structure Verification.

- Confirm that the main PHP/application entry file is present.
- Confirm that CSS, JavaScript, images/assets and supporting folders are present.
- Confirm that database connection/configuration files are present.
- Confirm that files used for authentication, appointments, payments, reports, feedback and other modules are present.
- If the project uses external libraries or APIs, confirm that their configuration files are included.

Permissions.

On Windows, the application normally runs from the XAMPP web directory without additional file-permission changes. If the project is deployed on Linux, configure the web-server user and directory permissions according to the server environment before starting the application.

7. Configuration.

Database Connection Configuration.

1. Open the PHP database connection/configuration file supplied with the project.
2. Set the database host to the local MySQL server, normally 127.0.0.1 or localhost, according to the existing project configuration.
3. Set the database username and password required by the local MySQL installation.
4. Set the database name to the exact name created/imported in phpMyAdmin.
5. Save the file and keep a secure backup of the original configuration.

Application and API Configuration.

- Configure email notification settings if the project package requires them.
- Configure API settings if enabled in the delivered application.

- Configure PayPal/payment integration credentials if payment functionality is being tested.
- Configure the AI-powered symptom checker/API integration if the delivered application requires an external service.
- Use test/sandbox credentials for demonstration where the external provider supports them.
- Localhost Configuration:
- Application address: <http://127.0.0.1/>
- Alternative local address: <http://localhost/>

Port and Service Check.

- Confirm Apache is running and serving PHP pages.
- Confirm MySQL is running and accessible through phpMyAdmin.
- If Apache or MySQL cannot start, identify and stop conflicting services before continuing.
- If a non-default Apache port is used, update the browser URL accordingly.

Security Configuration.

- Do not expose database passwords or API secrets in screenshots or public repositories.
- Use strong credentials for administrator accounts.
- The project report specifies role-based access, secure password handling and protection of sensitive areas such as billing and medical history.
- For production deployment, HTTPS should be configured even though the local development environment uses HTTP.

8. Starting the Application.

Starting the Web Server.

1. Open the XAMPP Control Panel.
2. Start Apache.
3. Start MySQL.
4. Wait until both services show that they are running.

Accessing the Application.

1. Open Google Chrome or Microsoft Edge.
2. Enter the localhost address for the project.
3. The HealthCare Plus home/login page should load.
4. Use the credentials supplied with the project for administrator, doctor, staff or patient testing.

Expected User Roles.

- Admin: Manages users, doctors, appointments, billing, staff and reports.
- Doctor: Accesses schedules, appointments and patient-related information permitted by the system.
- Patient: Registers, logs in, books or cancels appointments, makes payments and views permitted personal records.

Stopping the Application.

- Close the browser when testing is complete.
- Stop Apache and MySQL from the XAMPP Control Panel when the local environment is no longer required.

- Do not terminate XAMPP services while a database operation is actively running.

9. Verification and Testing.

Web Server Testing.

- Open the application URL in Chrome or Edge.
- Confirm that the home/login page loads without PHP errors.
- Navigate between the main pages and confirm that CSS and JavaScript resources load correctly.

Authentication Testing.

- Open the login page.
- Test a valid administrator account if provided.
- Test a valid patient, doctor or staff account if provided.
- Confirm that the appropriate role-based dashboard is displayed.
- Test an invalid login and confirm that an appropriate error message is shown.

Database Testing.

- Open phpMyAdmin and confirm that the project database exists.
- Check that the expected tables are present.
- Create or update a test record through the application.
- Refresh the relevant database table and verify that the change is stored.

Functional Testing.

1. Register a new patient account, if registration is enabled in the delivered build.
2. Log in using the new account.

3. Search for a doctor and view availability.
4. Book an appointment.
5. Confirm that the appointment appears in the booking/history area.
6. Test the payment workflow using approved test credentials where available.
7. Check receipt/payment information.
8. Test feedback submission.
9. Test the AI-powered symptom checker if its external service is configured.
10. Log in as an administrator and verify relevant records and reports.

10. Troubleshooting.

Problem	Possible Cause	Action
Apache will not start	Port conflict or another web service is running.	Stop the conflicting service or configure a different Apache port and update the URL.
MySQL will not start	Port/service conflict or damaged service state.	Check the XAMPP log, stop conflicting database services and restart MySQL.
phpMyAdmin cannot connect	MySQL is not running or credentials are incorrect.	Start MySQL and verify the configured credentials.
PHP page shows an error	Missing file, PHP error or incorrect configuration.	Check the Apache/PHP error log and verify the project files and configuration.

Database connection error	Wrong host, username, password or database name.	Compare the PHP connection settings with the database created in phpMyAdmin.
CSS/JS not loading	Incorrect relative paths or incomplete project copy.	Check the project folder structure and browser developer console.
Payment/API feature fails	Missing/incorrect third-party credentials or internet issue.	Check API configuration, test credentials and browser/server logs.
Symptom checker fails	AI/API configuration or network issue.	Verify the configured AI service/API key and internet connectivity.

Table 2: Troubleshooting.

General Recovery Procedure.

1. Stop Apache and MySQL.
2. Check the XAMPP Control Panel and Apache/MySQL logs.
3. Verify that the project files were copied completely.
4. Verify the PHP database connection settings.
5. Verify that the required database exists and contains the expected tables.
6. Restart MySQL and Apache.
7. Clear the browser cache or test in a private/incognito window.
8. Repeat the verification tests.

11.Backup and Maintenance.

Database Backup.

1. Open phpMyAdmin.
2. Select the HealthCare Plus database.
3. Choose the Export function.
4. Use the SQL format and export the complete database.
5. Store the backup in a secure location separate from the application folder.
6. Record the backup date and database version.

Application Backup.

- Keep a copy of the complete project source directory.
- Keep configuration values and deployment notes in a secure location.
- Do not include private API keys or passwords in publicly shared backups.
- Before modifying the database schema or PHP connection settings, create a backup.

Routine Maintenance.

- Review application and Apache/PHP logs for repeated errors.
- Remove obsolete test accounts and demonstration data before real deployment.
- Verify that database backups can be restored in a test environment.
- Update XAMPP/PHP/MySQL only after confirming compatibility with the project.
- Review API credentials and third-party service availability.
- Restrict access to administrative functions and sensitive hospital data.

12. Uninstallation.

Stopping Application Service.

- Close the HealthCare Plus application in the browser.
- Stop Apache in the XAMPP Control Panel.
- Stop MySQL in the XAMPP Control Panel.

Removing the Database.

1. Open phpMyAdmin while MySQL is running if the database needs to be removed through the interface.
2. Select the HealthCare Plus database.
3. Create a final backup if the data may be required later.
4. Use the database drop/delete operation only after confirming that the data is no longer required.
5. Stop MySQL after the database has been removed.

Removing Project Files.

- Open the XAMPP htdocs directory.
- Delete the HealthCare Plus project folder after confirming that a backup is available.
- Remove any project-specific configuration copies or temporary files.

Optional Software Removal.

- Uninstall XAMPP if it is no longer required for other projects.
- Uninstall Visual Studio Code or Notepad++ only if they are no longer required.
- Keep browser software installed if it is used for other systems.